



MatriWatch



Smart Material Health Monitoring Software For Mothers

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1 LITERATURE REVIEW

Postpartum depression affects mothers, infants, and family well-being, but early detection is still limited in low-resource settings. Tools like PHQ-9 and EPDS are widely used, while machine learning can improve fast and data-driven screening.

2 RESEARCH OBJECTIVE

- Build a Bangladesh-focused postpartum depression screening model.
- Predict PHQ-9 severity and EPDS risk level.
- Compare Attention-based 1D-CNN and RBF-SVM.
- Handle missing values and class imbalance.
- Explain predictions using SHAP and feature importance.

4 METHODOLOGY

The dataset of 800 postpartum women was cleaned, encoded, standardized, and balanced using SMOTE. Two models, Attention-based 1D-CNN and RBF-SVM, were trained and compared. The best model was selected using validation performance and tested using accuracy, weighted F1-score, ROC curves, confusion matrix, and SHAP analysis.

Methodology Pipeline: Score-Free Task-Adaptive Hybrid PPD Framework

Raw PHQ-9 / EPDS total scores are excluded from the predictive features throughout the main pipeline.

1. Data preparation

Raw Survey Data (800)

Score-Free Preprocessing & Encoding

SMOTE Balancing

Train / Val / Test Split 70 / 15 / 15

2. Task-adaptive modeling and outputs

Attention-Based 1D-CNN Branch

Conv1D → BN → ReLU

Channel attention + FC head

Tuned RBF-SVM Decision Branch

PHQ-9: C=2, gamma=0.1

EPDS: C=3, gamma=0.05

Validation-Based Task Selector

Branch with higher validation weighted F1 wins

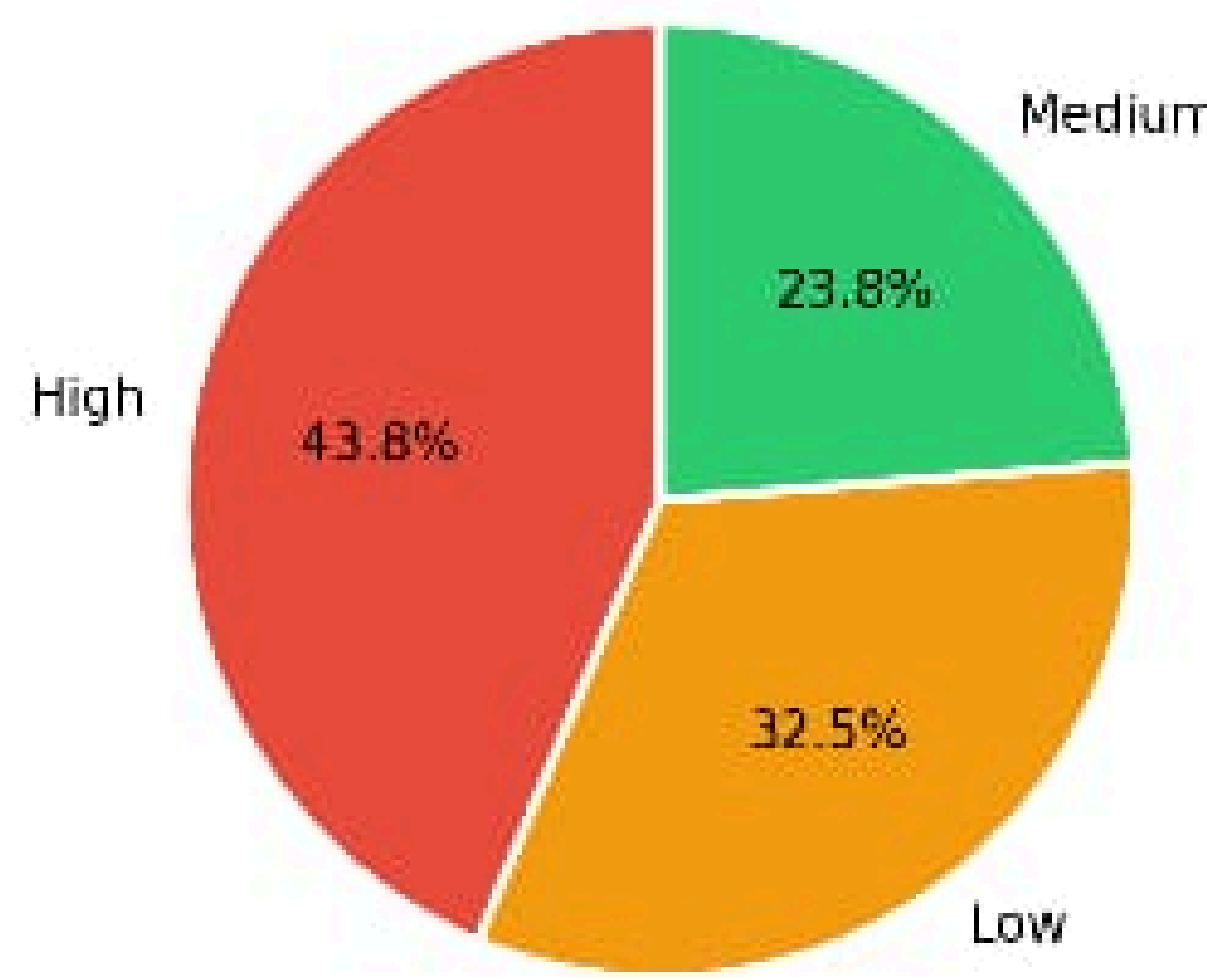
PHQ-9 Final 6-class Output

Score-free severity classification

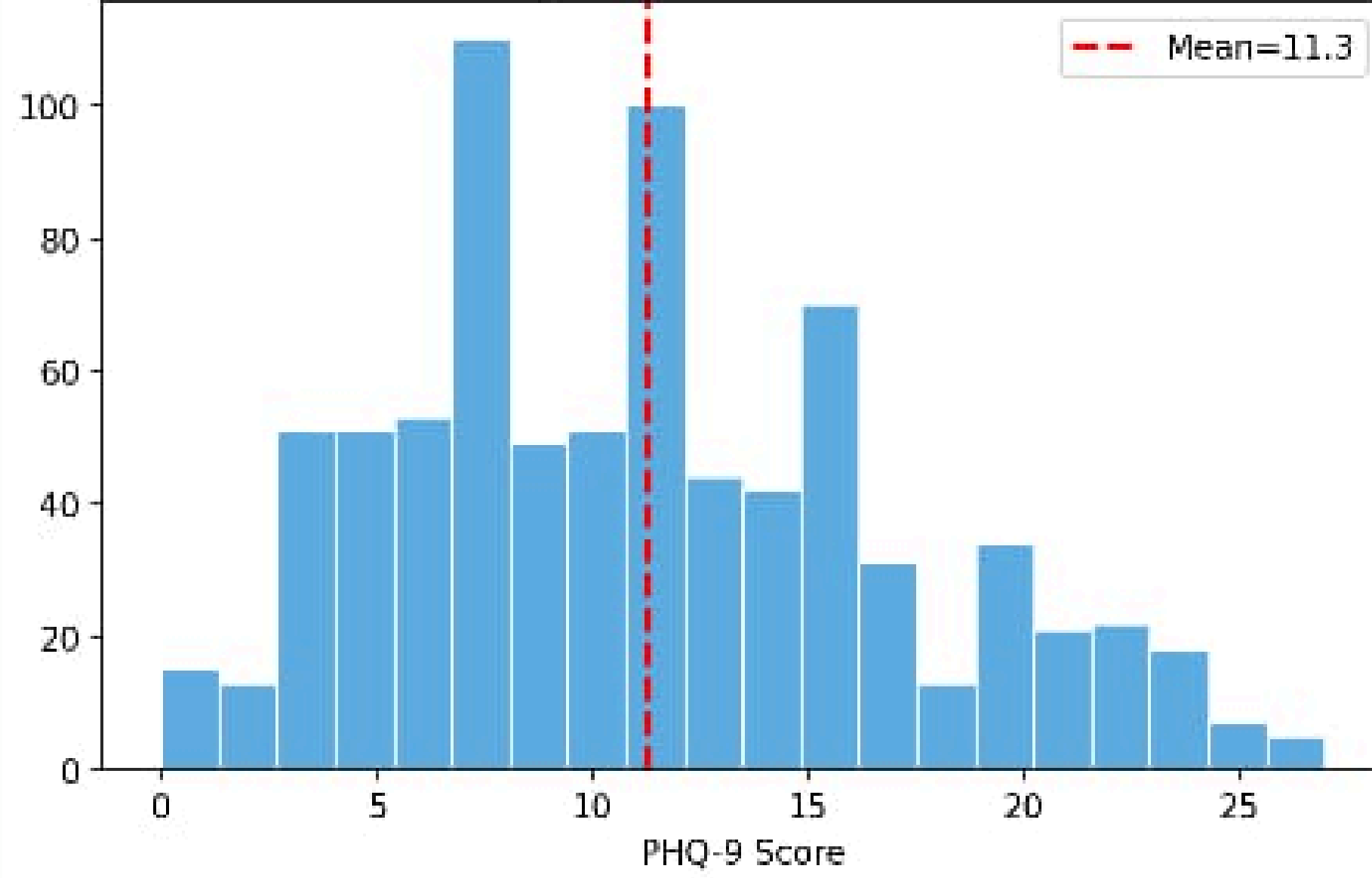
EPDS Final 3-class Output

Score-free risk classification

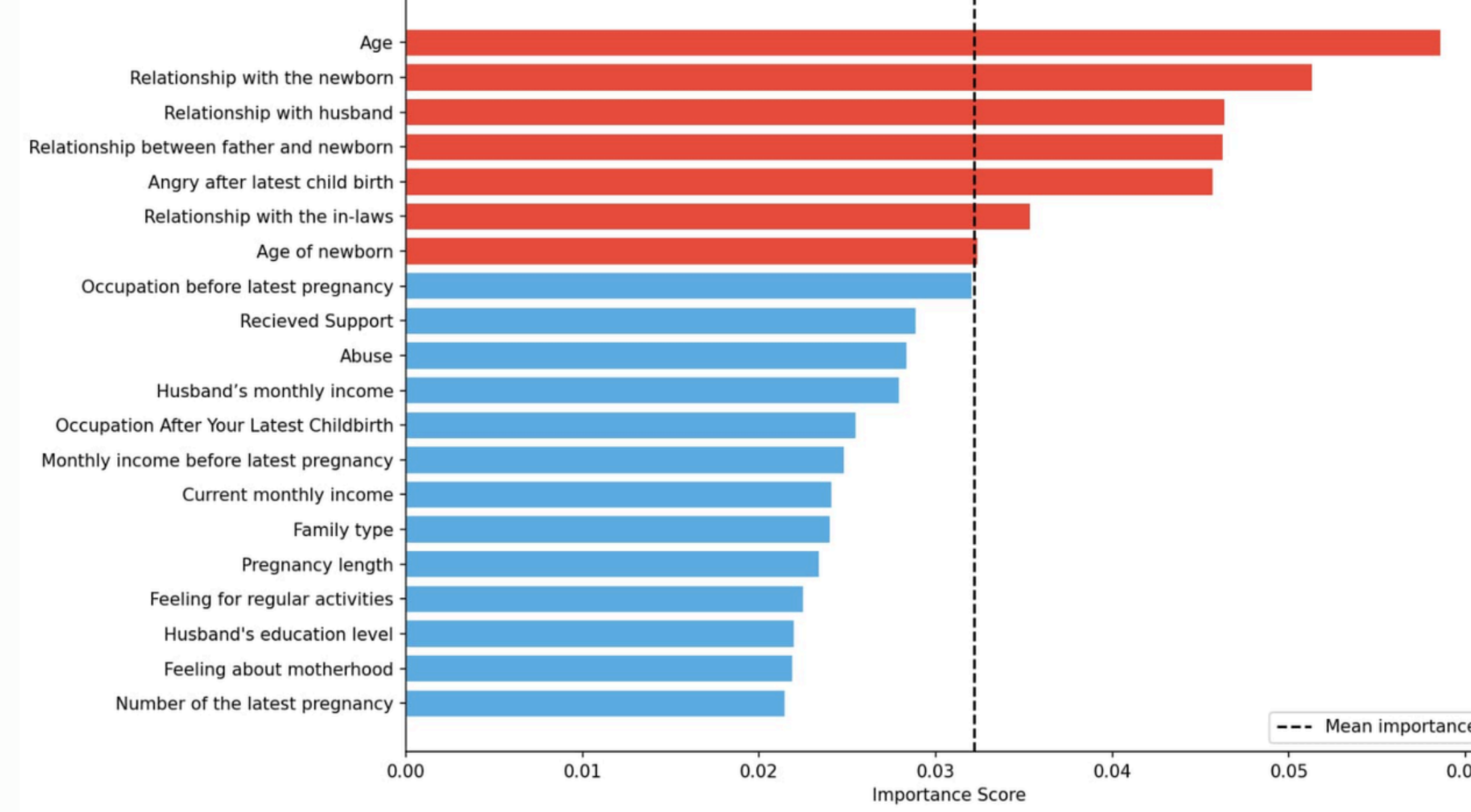
EPDS Severity Distribution



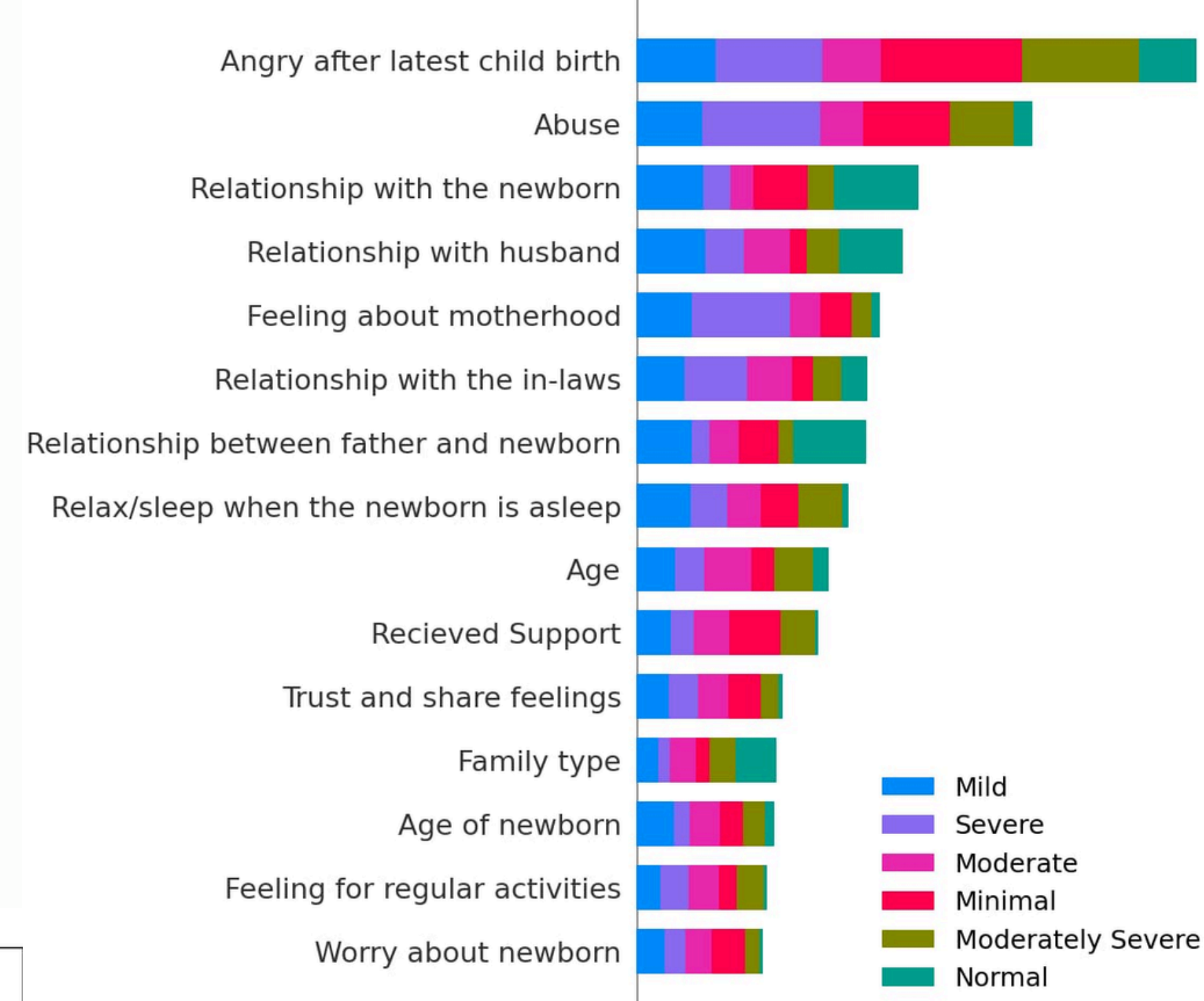
PHQ-9 Score Distribution



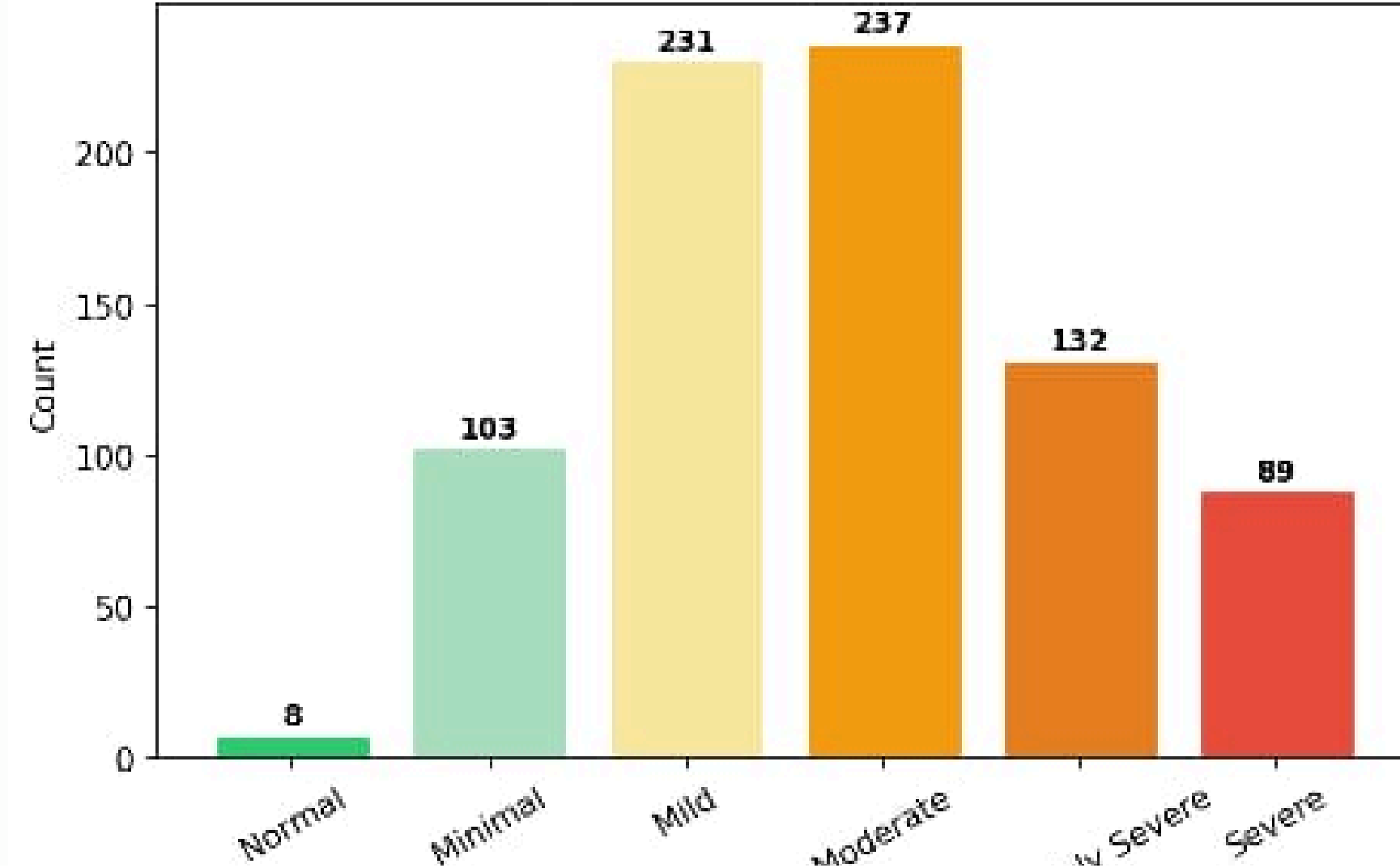
Top 20 Feature Importances (Score-free Random Forest) - PHQ-9 Target



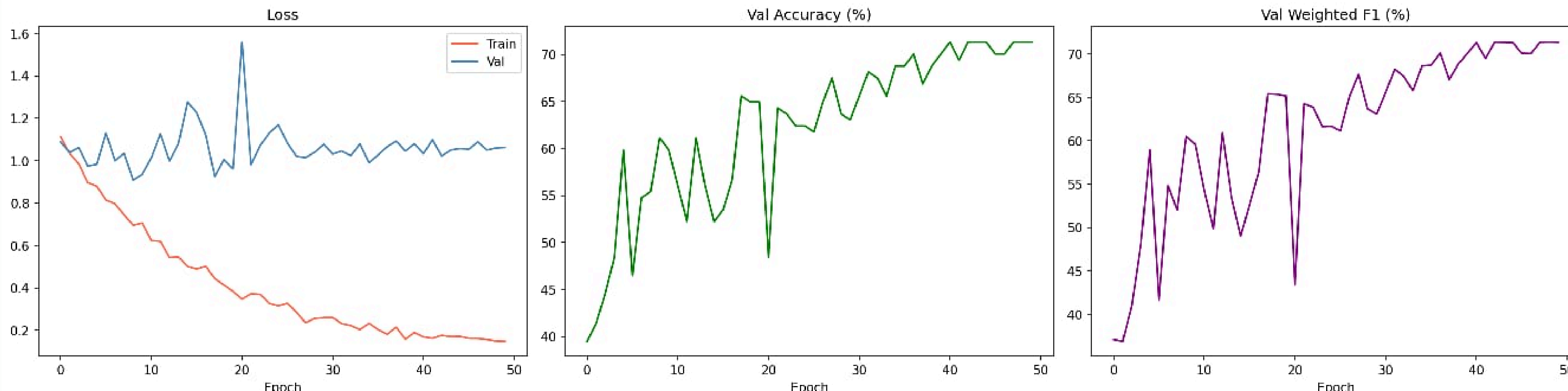
SHAP Summary - PHQ-9 Multiclass (Score-free RF)



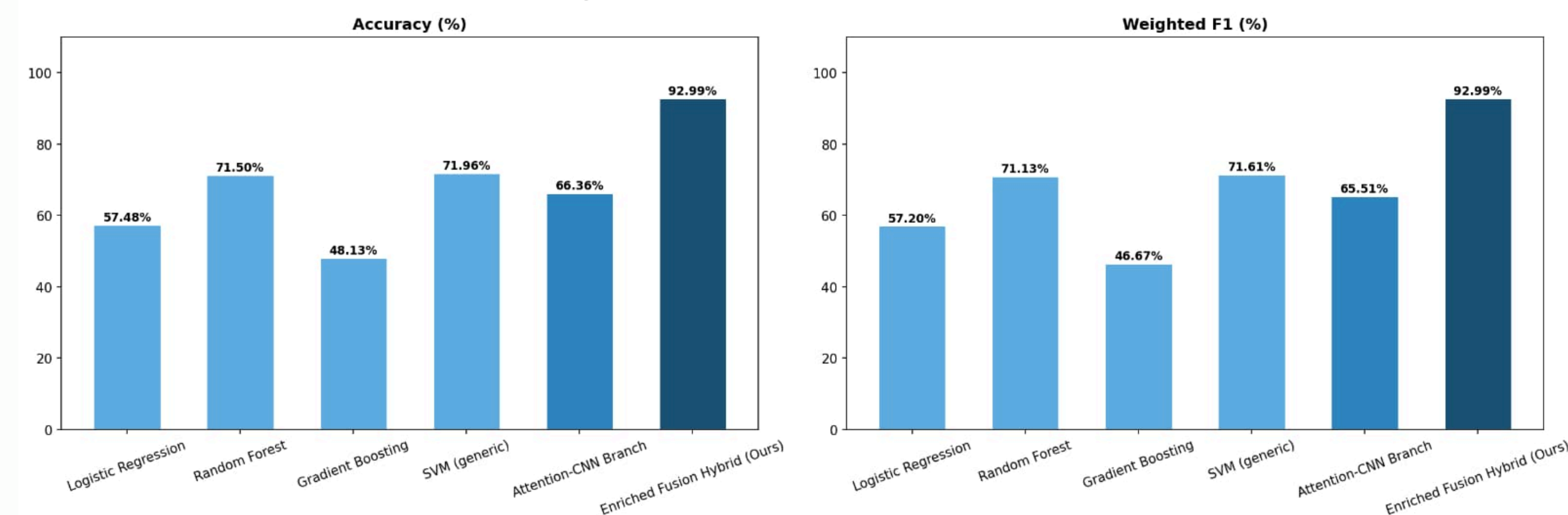
PHQ-9 Severity Distribution



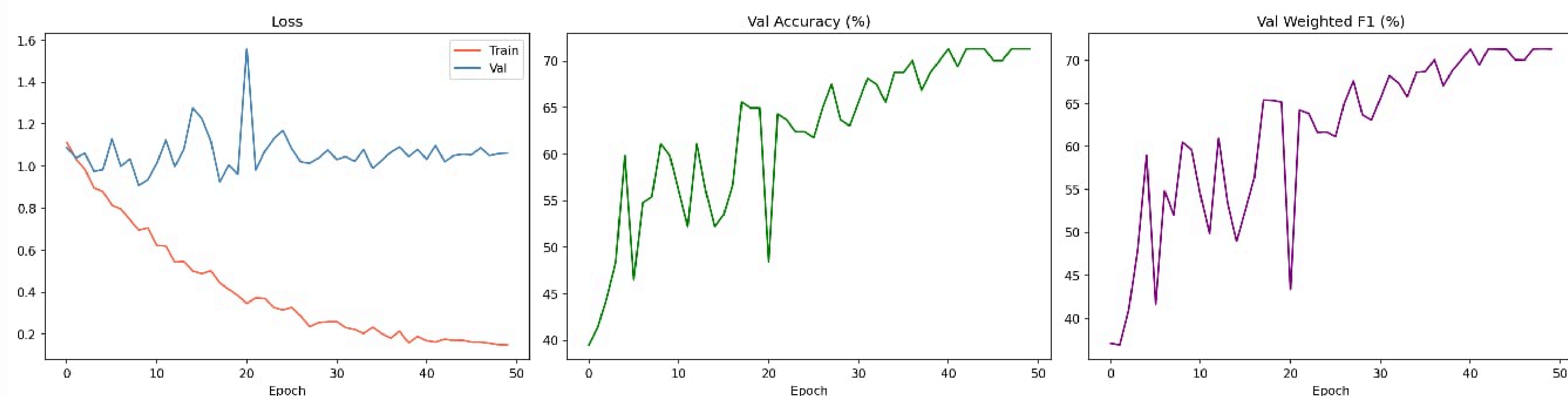
EPDS Model - Training Curves



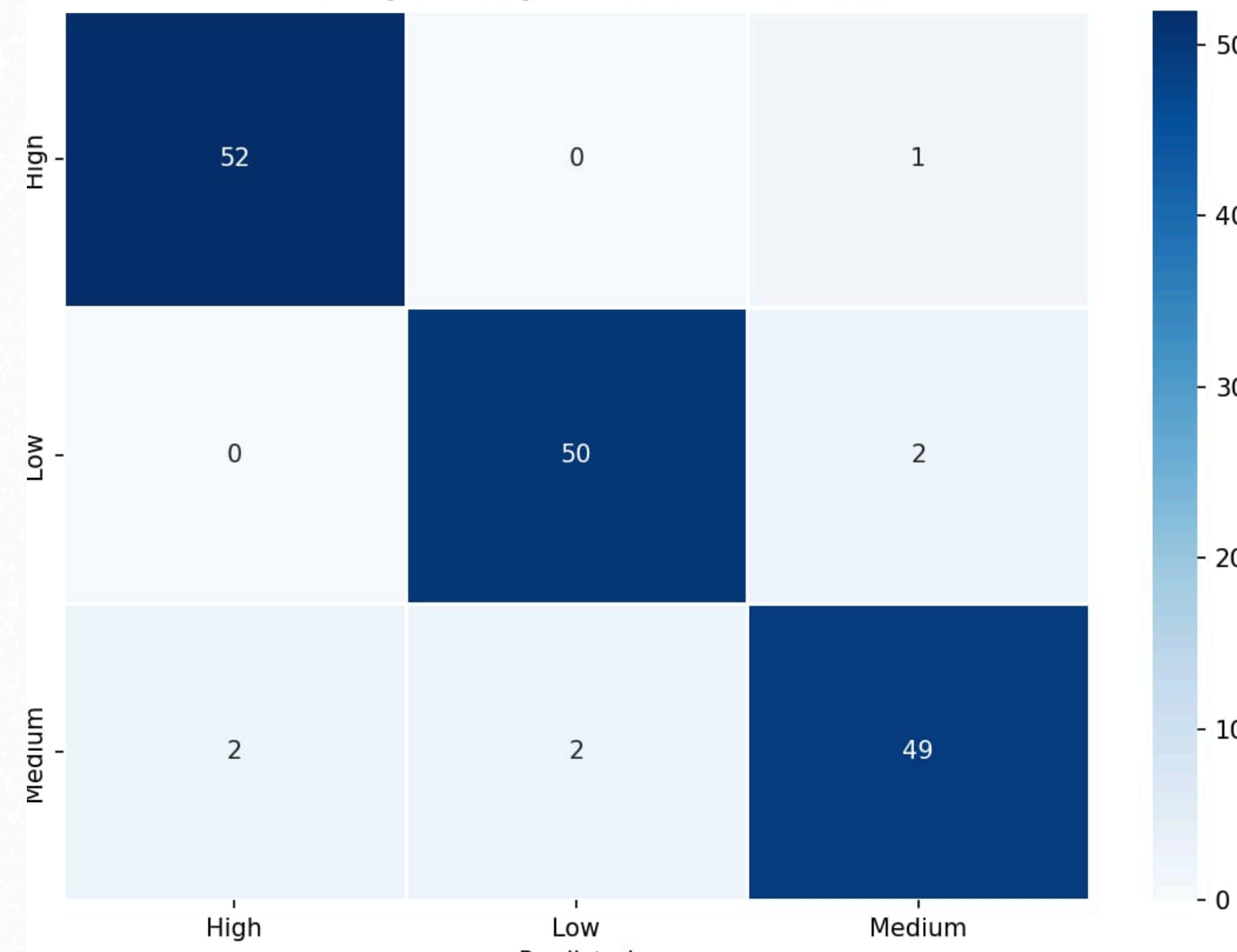
PHQ-9 Comparison - Score-free Baselines vs Enriched Fusion Final



EPDS Model - Training Curves



EPDS (3-class) - Confusion Matrix



5 CONCLUSION

The proposed score-free model achieved strong performance for both PHQ-9 and EPDS screening. Results show that psychosocial, relational, abuse-related, newborn, and postpartum factors can support early and interpretable depression risk detection in low-resource maternal healthcare settings.